

Measurements & Errors

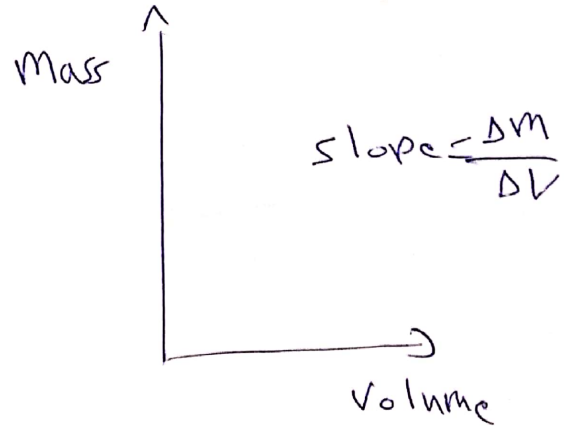
التجربة الأولى

$$D = \frac{m}{V}$$

$$V_{\text{cylinder}} = \pi r^2 L$$

$$\text{slope} = \frac{\Delta m}{\Delta V} = D = \frac{\Delta m}{\frac{\pi D^2 h}{4}}$$

$$\frac{\Delta D}{D} = \frac{\Delta m}{m} + \frac{2\Delta D}{D} + \frac{\Delta h}{h}$$



Projectile Motion

التجربة الثانية

$$V_0 \text{ theory} = \sqrt{\frac{10}{7} g y}$$

$$h = \frac{g D^2}{2 V_0^2}$$

$$\text{slope} = \frac{\Delta h}{D^2} = \frac{g D^2}{2 V_0^2 D^2}$$

$$\text{slope} = \frac{g}{2 V_0^2}$$

$$g = \text{slope} \times 2 V_0^2$$



$$\frac{\Delta g}{g} = \frac{\Delta h}{h} + \frac{2\Delta D}{D} + \frac{\Delta g}{g}$$

Newton's 2nd law

Atwood machine

Part 1

constant mass

$$g_{\text{theory}} = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

$$g_{\text{exp}} = \frac{2y}{t^2}$$

g_{exp}

$$\text{slope} = \frac{\Delta g}{\Delta(m_2 - m_1)}$$

$m_2 - m_1$

$$\text{slope} = \frac{\Delta g_{\text{exp}}}{\Delta(m_2 - m_1)} = \frac{g}{m_1 + m_2}$$

$$g = \text{slope} (m_1 + m_2)$$

$$\frac{\Delta g_{\text{exp}}}{g_{\text{exp}}} \cdot \frac{\Delta g}{g} = \frac{\Delta y}{y} + \frac{2\Delta t}{t}$$

$$\text{Percent error} \Rightarrow \frac{\Delta g}{g} \times 100\%$$

Part 2

constant net Force

$$\text{slope} = \frac{g_{\text{exp}}}{\frac{1}{m_1 + m_2}}$$

$$\text{slope} = (m_2 - m_1)g$$

$$g = \frac{\text{slope}}{m_2 - m_1}$$

$g(\text{exp})$

$$\text{slope} = \frac{\Delta g_{\text{exp}}}{\frac{1}{m_1 + m_2}}$$

$\frac{1}{m_1 + m_2}$

$$\frac{\Delta g_{\text{exp}}}{g_{\text{exp}}} = \frac{\Delta g}{g} = \frac{\Delta y}{y} + \frac{2\Delta t}{t}$$

$\frac{1}{m_1 + m_2}$

x_{max}

x_{et}

x_{et}

Conservation of Energy

$$E = K + U \rightarrow mgh$$

$$\frac{1}{2}mv^2 \rightarrow \frac{1}{2}kx^2$$

$$X = h = mgt^2$$

$$2(m+M)$$

$$\text{slope} = \frac{\Delta x}{\Delta t^2} = \frac{mg}{2(m+M)}$$

$$g = \frac{2(m+M) \times \text{slope}}{m}$$

$$\frac{\Delta g}{g} = \frac{\Delta x}{x}$$

part 2 1. spring constant

$$F = -kx$$

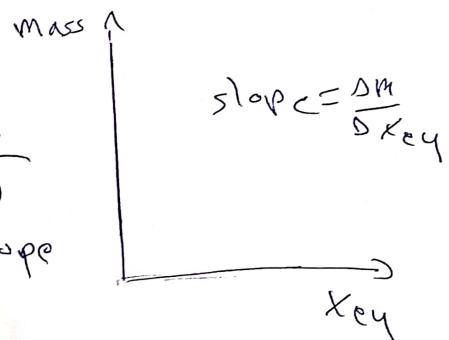
$$kx = mg$$

$$\frac{\Delta k}{k} = \frac{\Delta x_{eq}}{x_{eq}}$$

$$\text{slope} = \frac{\Delta m}{\Delta x} = \frac{k}{g}$$

$$k = g \times \text{slope}$$

a.e. m/s^2



2. spring using mass - system

$$\text{slope} = \frac{\Delta m}{\Delta x} = \frac{1}{2} \frac{k}{g}$$

$$g = \frac{k}{2 \text{slope}}$$

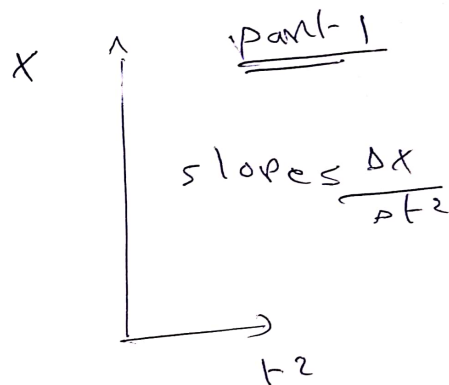
$$\frac{\Delta g}{g} = \frac{\Delta k}{k} + \frac{\Delta x_{max}}{x_{max}} = \frac{\Delta x_{eq}}{x_{eq}} + \frac{\Delta x_{max}}{x_{max}}$$

التجربة الثانية

$$F = \square H^2$$

$$T = \frac{1}{F}$$

النصف الثاني من التجربة



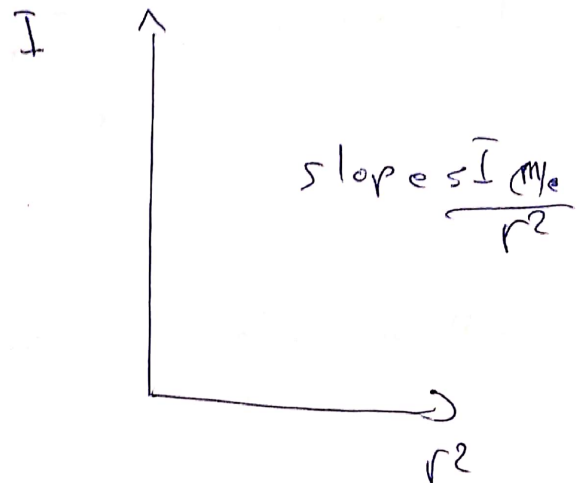
Rotational dynamics

$$I = \frac{Mgd^2 t^2}{8g}$$

$$I_{\text{rod}}^{\text{theory}} = \frac{1}{12} mL^2$$

$$I_{\text{rod}}^{\text{exp}} = \text{تقدير العزم الدوراني}$$

التجربة الخالية



$$\frac{\Delta I_{\text{rod}}}{I_{\text{rod}}} = \frac{\Delta m}{m} + \frac{2\Delta L}{L}$$

$$I_{\text{rod}} = \frac{1}{12} mL^2$$

$$\frac{\Delta I}{I} = \frac{\Delta g}{g} + \frac{2\Delta t}{t}$$

$$I = \frac{Mgd^2 t^2}{8g}$$

Ohm's Law

التجربة الثانية

$$R = \frac{V}{I}$$

$$R = \frac{\rho \times L}{A}$$

ρ ← المقاومة
 L ← طول السلك
 A ← مساحة المقطع العرضي للسلك

① connection of Resistors R_1 & R_2

$$\text{slope 1} = \frac{\Delta V_1}{I_1}$$

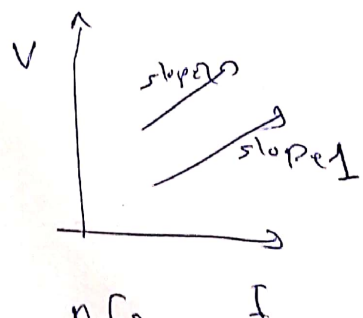
$$R_1 = \text{slope 1}$$

$$\text{slope 2} = \frac{\Delta V_2}{\Delta I_2}$$

$$R_2 = \text{slope 2}$$

$$\frac{\Delta R_1}{R_1} = \frac{\Delta V_1}{V_1} + \frac{\Delta I_1}{I_1}$$

$$\frac{\Delta R_2}{R_2} = \frac{\Delta V_2}{V_2} + \frac{\Delta I_2}{I_2}$$



2 connection of Resistors in series and parallel

- In series $R_{eq} = R_1 + R_2$

$$I_1 = I_2 = I$$

$$V = V_1 + V_2$$

- In parallel $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

$$V_1 = V_2 = V$$

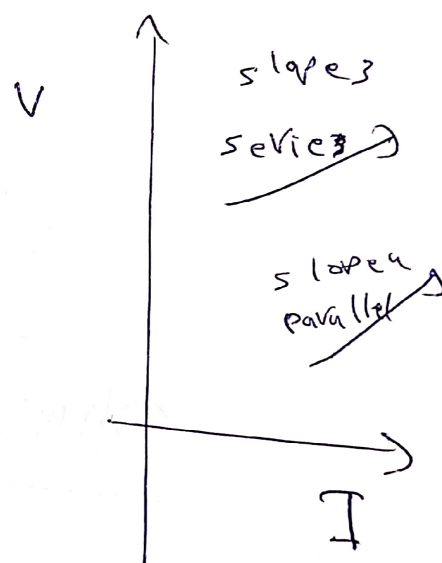
$$\cancel{V} = \cancel{V}$$
$$I = I_1 + I_2$$

R series = slope 3

R parallel = slope 4

$$\frac{\Delta R_{\text{series}}}{R_{\text{series}}} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R_{\text{parallel}}}{R_{\text{parallel}}} = \frac{\Delta V}{V} - \frac{\Delta I}{I}$$



Galvanometer

التيار الكهربائي

- By connect (G) with small Resistance in parallel

$$R_s = R_g I_g$$

$$\frac{\Delta I_d}{I_d} = \frac{\Delta d}{d} + \frac{\Delta L}{L} \quad I_{max} = I_g$$

- By connect (V) with large Resistance in series

$$R_m = \frac{V_{max}}{I_g} = R_g I_s$$

$$\frac{\Delta V_d}{V_d} = \frac{\Delta d}{d} + \frac{\Delta R_m}{R_m}$$

The potentiometer

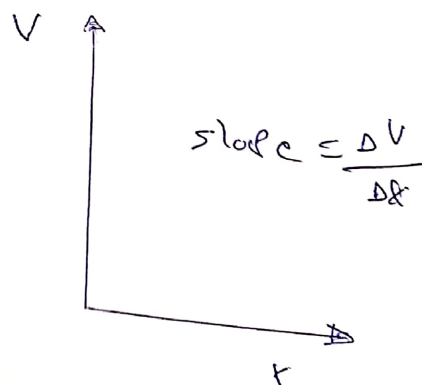
الدوائر الكهربائية

$$\frac{V_1}{V_2} = \frac{L_1}{L_2}$$

Part 1

calibration curve

$$\text{slope} = \frac{\Delta V}{\Delta x} = \frac{I \rho}{A}$$



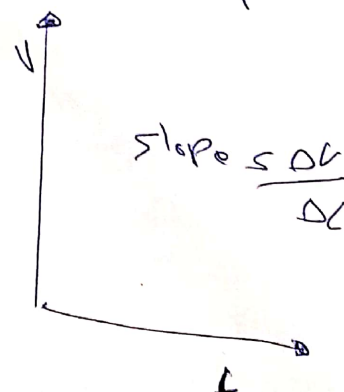
Part 2

E. M. F

$$\text{slope} = \frac{E_x}{L_x} = \frac{E_x}{L_x}$$

$$E_x = L_x \times \text{slope}$$

$$\frac{\Delta E_x}{E_x} = \frac{\Delta L_x}{L_x} + \frac{\Delta V}{V}$$



RC circuit

التجربة الثانية

$$Q = CV$$

$$T = RC \rightarrow \text{الزمن الذي يستغرقه الشحن أو التفريغ}$$

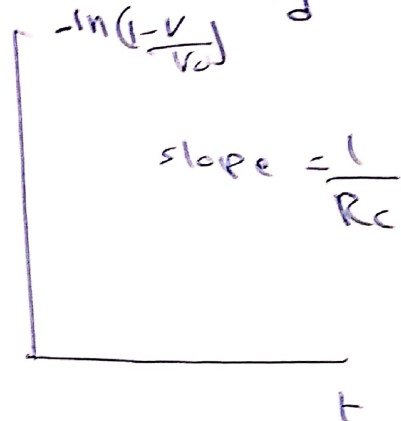
= charging of a capacitor

$$V = V_0 \left(1 - e^{-t/RC}\right)$$

$$-\ln\left(1 - \frac{V}{V_0}\right) = \frac{t}{RC}$$

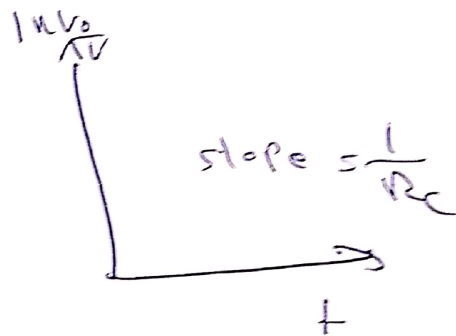
$$C = \frac{1}{R \times \text{slope}}$$

$$C = \frac{EA}{d}$$



= Discharging

$$\frac{\ln V_0}{V} = \frac{t}{RC}$$



$$R = \frac{1}{\text{slope}} \rightarrow \text{From part 1}$$

Wheatstone Bridge

التجربة الثالثة

$$R_x = R_1 \frac{R_3}{R_2}$$

$$R_x = R_1 \frac{b}{a}$$

$$b = \text{length of wire}$$

$$a = 100 - b$$

part 1 $\Rightarrow R_{x1, \text{exp}} = R_1 \frac{b}{a}$

$R_{\text{theory}} = \text{calculated value}$

part 3

$$R_{\text{theory}} = R_1 \frac{b}{a}$$

$$R_{\text{theory}} = R_{x1} + R_{x2}$$

part 2 $\Rightarrow R_{x2, \text{exp}} = R_1 \frac{b}{a}$

R_{theory}

part 4 $\Rightarrow \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

$$R_{\text{theory}} = R_1 \frac{b}{a}$$

$$R_{\text{theory}} = \frac{R_{x1} R_{x2}}{R_{x1} + R_{x2}}$$